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LIQUID EJECTING HEAD

Abstract

A liquid ejection head includes: a vibration plate, and a piezoelectric element having a first electrode, a piezoelectric layer, and a second electrode, which are laminated in a lamination direction, wherein the piezoelectric layer is formed of a piezoelectric material containing lead as a constituent element, the first electrode has an electrode layer and a lead-containing layer disposed between the electrode layer and the vibration plate and containing lead, and when, of two regions arranged in a crossing direction, which is a direction crossing the lamination direction, when seen in the lamination direction, a region farther away from a center of the first electrode is a first region and a region closer to the center of the first electrode is a second region, a lead content percentage in the lead-containing layer in the first region is higher than a lead content percentage in the lead-containing layer in the second region.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-031939, filed Mar. 4, 2024 and JP Application Serial Number 2024-170142, filed Sep. 30, 2024, the disclosures of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a liquid ejection head.

2. Related Art

[0003] A liquid ejection apparatus, typified by a piezoelectric ink jet printer, uses a liquid ejection head that ejects liquid such as ink. For example, the head described in JP-A-2002-319714 has a vibration plate and a piezoelectric element. The piezoelectric element is formed by a lower electrode, a piezoelectric film, and an upper electrode laminated sequentially on the vibration plate.

[0004] In the head described in JP-A-2002-319714, depending on a lead content in the lower electrode, stress generated in the lower electrode may be greatly concentrated, which may result in damage such as cracking.

SUMMARY

[0005] A liquid ejection head includes: a piezoelectric element having a first electrode, a piezoelectric layer, and a second electrode and a vibration plate, the first electrode, the piezoelectric layer, and the second electrode are laminated in this order in a lamination direction which is a direction directed from the vibration plate to the piezoelectric element, the piezoelectric layer is formed of a piezoelectric material containing lead as a constituent element, the first electrode has an electrode layer and a lead-containing layer which is disposed between the electrode layer and the vibration plate and contains lead, and when, of two regions arranged in a crossing direction, which is a direction crossing the lamination direction, when seen in the lamination direction, a region farther away from a center of the first electrode is a first region and a region closer to the center of the first electrode is a second region, a first lead content percentage in the lead-containing layer in the first region is higher than a second lead content percentage in the lead-containing layer in the second region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a configuration diagram schematically showing a liquid ejection

[0007] apparatus having a liquid ejection head according to an embodiment.

[0008] FIG. **2** is an exploded perspective view of the liquid ejection head according to the embodiment.

[0009] FIG. **3** is a sectional view taken along the III-III line in FIG. **2**.

[0010] FIG. **4** is a plan view showing part of the liquid ejection head according to the embodiment.

[0011] FIG. **5** is a sectional view taken along the V-V line in FIG. **4**.

[0012] FIG. **6** is an enlarged view of part VI in FIG. **5**.

[0013] FIG. **7** is a diagram showing the relation between lead content information on a lead-containing layer and evaluations of liquid ejection heads.

DESCRIPTION OF EMBODIMENTS

[0014] Preferred embodiments according to the present disclosure are described below with reference to the drawings attached hereto. Note that the dimensions and scales of components in the drawings differ from the actual ones as needed, and some portions are depicted schematically in order to facilitate understanding. Also, the scope of the present disclosure is not limited to those modes of the present disclosure unless the following description specifically states so.

[0015] Note that the following description uses an X-axis, a Y-axis, and a Z-axis that are orthogonal to one another, as needed. Also, in the following description, one direction along the X-axis is an X1-direction, and a direction opposite from the X1-direction is an X2-direction. Similarly, directions along the Y-axis that are opposite from each other are a Y1-direction and a Y2-direction. Also, directions along the Z-axis that are opposite from each other are a Z1-direction and a Z2-direction. The Z1-direction is an example of the “lamination direction.” Also, a view seen in a direction along the Z-axis may be referred to as a “plan view”.

[0016] The Z-axis is typically a vertical axis, and the Z2-direction corresponds to a vertically downward direction. However, the Z-axis does not have to be a vertical axis.

[0017] Also, although the X-axis, the Y-axis, and the Z-axis are typically orthogonal to one another, the present disclosure is not limited to this as long as they intersect at an angle in the range of, for example, no less than 80° and no more than 100°.

1. Embodiment

1-1. Overall Configuration of a Liquid Ejection Apparatus

[0018] FIG. **1** is a configuration diagram schematically showing a liquid ejection apparatus **100** having a liquid ejection head **50** according to an embodiment. The liquid ejection apparatus **100** is an ink jet printing apparatus that ejects droplets of ink, which is an example of the liquid, to a medium M. The medium M is typically a print sheet. Note that the medium M is not limited to a print sheet, and any material, such as, e.g., a resin film or a cloth, may be used for printing.

[0019] As shown in FIG. **1**, the liquid ejection apparatus **100** has a liquid container **10**, a control unit **20**, a conveyance mechanism **30**, a movement mechanism **40**, and the liquid ejection head **50**.

[0020] The liquid container **10** is a container storing ink. Example specific modes of the liquid container **10** include a cartridge attachable to and detachable from the liquid ejection apparatus **100**, a bag-shaped ink pack formed of a flexible film, and an ink tank that can be replenished with ink. Note that the type of ink stored in the liquid container **10** is not limited to any particular type and may be any type.

[0021] The control unit **20** includes, for example, a processing circuit such as a central processing unit (CPU) or a field-programmable gate array (FPGA) and a memory circuit such as semiconductor memory, and controls the operation of each component in the liquid ejection apparatus **100**.

[0022] As controlled by the control unit **20**, the conveyance mechanism **30** conveys the medium M in the Y2-direction. As controlled by the control unit **20**, the movement mechanism **40** causes the liquid ejection head **50** to reciprocate in the X1-direction and the X2-direction. In the example shown in FIG. **1**, the movement mechanism **40** has a substantially box-shaped carriage **41** housing the liquid ejection head **50** and a conveyor belt **42** to which the carriage **41** is fixed. Note that the number of liquid ejection heads **50** mounted to the carriage **41** is not limited to one and may be more than one. Also, in addition to the liquid ejection head **50**, the liquid container **10** described earlier may be mounted to the carriage **41**.

[0023] As controlled by the control unit **20**, the liquid ejection head **50** ejects ink supplied from the liquid container **10** to the medium M in the Z2-direction from each of a plurality of nozzles. This ejection is performed in concurrence with the conveyance of the medium M by the conveyance mechanism **30** and the reciprocating movement of the liquid ejection head **50** by the movement mechanism **40**, thereby forming an image on the surface of the medium M with the ink.

1-2. Overall Configuration of the Liquid Ejection Head

[0024] FIG. **2** is an exploded perspective view of the liquid ejection head **50** according to the embodiment. FIG. **3** is a sectional view taken along the III-III line in FIG. **2**. As shown in FIGS. **2** and **3**, the liquid ejection head **50** has a flow channel substrate **51**, a pressure chamber substrate **52**, a nozzle plate **53**, a vibration absorber **54**, a vibration plate **55**, a plurality of piezoelectric elements **56**, a seal plate **57**, a case **58**, and a wiring substrate **59**. The vibration plate **55** and the piezoelectric element **56** form an actuator **1**. In this way, the liquid ejection head **50** has the actuator **1** including the piezoelectric element **56** and the vibration plate **55**.

[0025] The pressure chamber substrate **52**, the vibration plate **55**, the plurality of piezoelectric elements **56**, the case **58**, and the seal plate **57** are disposed in a region located on the Z1-direction side of the flow channel substrate **51**. Meanwhile, the nozzle plate **53** and the vibration absorber **54** are disposed in a region located on the Z2-direction side of the flow channel substrate **51**. The components of the liquid ejection head **50** are roughly plate members elongated in a direction along the Y-axis and are joined to each other using, for example, an adhesive.

[0026] As shown in FIG. **2**, the nozzle plate **53** is a plate-shaped member provided with a plurality of nozzles N arrayed in a direction along the Y-axis. Each nozzle Nis a through-hole through which ink passes. In this way, the nozzle plate **53** has the plurality of nozzles N that produce jets of ink. The nozzle plate **53** is manufactured by, for example, processing a single-crystal silicon substrate using a semiconductor manufacturing technique employing a processing technique such as dry etching or wet etching. However, other publicly known methods and materials may be used as needed to manufacture the nozzle plate **53**.

[0027] The flow channel substrate **51** is a plate-shaped member forming flow channels of ink. As shown in FIGS. **2** and **3**, the flow channel substrate **51** is provided with an opening portion R**1**, a plurality of supply flow channels Ra, and a plurality of communication flow channels Na. The opening portion R**1** is an elongated through-hole extending in a direction along the Y-axis continuously over the plurality of nozzles N in a plan view seen in a direction along the Z-axis. Meanwhile, the supply flow channels Ra and the communication flow channels Na are through-holes provided for the individual nozzles N. The plurality of supply flow channels Ra each communicate with the opening portion R**1**. Similarly to the nozzle plate **53** described above, the flow channel substrate **51** is manufactured by, for example, processing a single-crystal silicon substrate using a semiconductor manufacturing technique. However, other publicly known methods and materials may be used as needed to manufacture the flow channel substrate **51**.

[0028] The pressure chamber substrate **52** is a plate-shaped member where a plurality of pressure chambers C are formed in correspondence to the plurality of nozzles N. The pressure chambers C are spaces called cavities, located between the flow channel substrate **51** and the vibration plate **55** and used to apply pressure to ink filled into the pressure chambers C. The plurality of pressure chambers C are arrayed in a direction along the Y-axis. Each pressure chamber C is formed by a hole **52***a* opening to both surfaces of the pressure chamber substrate **52** and has an elongated shape extending in a direction along the X-axis. In this way, the pressure chamber substrate **52** has the plurality of pressure chambers C communicating with the nozzles N. Each pressure chamber C, at its end on the X2-direction side, communicates with a corresponding one of the supply flow channels Ra. Meanwhile, each pressure chamber C, at its end on the X1-direction side, communicates with a corresponding one of the communication flow channel Na.

[0029] Similarly to the nozzle plate **53** described above, the pressure chamber substrate **52** is manufactured by, for example, processing a single-crystal silicon substrate using a semiconductor manufacturing technique. However, other publicly known methods and materials may be used as needed to manufacture the pressure chamber substrate **52**.

[0030] The vibration plate **55** is disposed at the surface of the pressure chamber substrate **52** which faces the Z1-direction. The vibration plate **55** is a plate-shaped member which is elastically deformable and is connected to the piezoelectric elements **56**. Note that details of the vibration plate **55** will be described later based on FIG. **5**.

[0031] The piezoelectric elements **56** are disposed at the surface of the vibration plate **55** which faces the Z1-direction. Each piezoelectric element **56** is a passive element that deforms upon supply of a drive signal and has an elongated shape extending in a direction along the X-axis. Note that one piezoelectric element **56** is provided for every pressure chamber, and the plurality of piezoelectric elements **56** are arrayed in a direction along the Y-axis in correspondence to the plurality of pressure chambers C. The vibration plate **55** vibrating in conjunction with deformation of the piezoelectric elements **56** fluctuates the pressure inside the pressure chambers C, causing the ink to be ejected from the nozzles N. Note that details of the piezoelectric elements **56** will be described later based on FIGS. **4** to **6**.

[0032] The case **58** is a case for storing ink to be supplied to the plurality of pressure chambers C and is joined, by means of an adhesive or the like, to the surface of the flow channel substrate **51** which faces the Z1-direction. The case **58** is formed by, for example, a resin material and is manufactured by injection molding. The case **58** is provided with a storage portion R**2** and an inlet IH. The storage portion R**2** is a concave portion whose outer shape conforms to the opening portion R**1** of the flow channel substrate **51**. The inlet IH is a through-hole communicating with the storage portion R**2**. The space formed by the opening portion R**1** and the storage portion R**2** functions as a liquid storage chamber R as a reservoir for storing ink. Ink from the liquid container **10** is supplied to the liquid storage chamber R through the inlet IH.

[0033] The vibration absorber **54** is a component for absorbing fluctuations in the pressure inside the liquid storage chamber R. The vibration absorber **54** is, for example, a compliance substrate which is a flexible, elastically deformable sheet member. The vibration absorber **54** is disposed at the surface of the flow channel substrate **51** which faces the Z2-direction, closing the opening portion R**1** and the plurality of supply flow channels Ra of the flow channel substrate **51** to form the bottom surface of the liquid storage chamber R.

[0034] The seal plate **57** is a structure that not only protects the plurality of piezoelectric elements **56** but also reinforces the mechanical strength of the pressure chamber substrate **52** and the vibration plate **55**. The seal plate **57** is joined to a surface of the vibration plate **55** using, for example, an adhesive. The seal plate **57** is provided with a concave portion for housing the plurality of piezoelectric elements **56**.

[0035] The wiring substrate **59** is joined to the surface of the pressure chamber substrate **52** or the vibration plate **55** which faces the Z1-direction. The wiring substrate **59** is a mount component where a plurality of interconnections are formed to electrically connect the control unit **20** and the liquid ejection head **50**. The wiring substrate **59** is, for example, a flexible wiring substrate such as a flexible printed circuit (FPC) or a flexible flat cable (FFC). A drive circuit **60** for driving the piezoelectric elements **56** is mounted on the wiring substrate **59**. The drive circuit **60** selectively supplies each piezoelectric element **56** with a drive signal for driving the piezoelectric element **56** via the wiring substrate **59**.

1-3. Details of the Vibration Plate and the Piezoelectric Elements

[0036] FIG. **4** is a plan view showing part of the liquid ejection head **50** according to the embodiment. FIG. **5** is a sectional view taken along the V-V line in FIG. **4**. Based on FIGS. **4** and **5**, the following describes the pressure chamber substrate **52**, the piezoelectric element **56**, and the vibration plate **55** in this order.

[0037] As shown in FIGS. **4** and **5**, the pressure chamber substrate **52** is provided with the holes **52***a* forming the pressure chambers C. The pressure chamber substrate **52** is accordingly provided with a wall-shaped partitioning wall **52***b* between each pair of adjacent holes **52***a*, the partitioning wall **52***b* extending in a direction along the X-axis. The pressure chamber substrate **52** is manufactured by, for example, processing a single-crystal silicon substrate using a semiconductor manufacturing technique. FIG. **4** shows, using broken lines, the plan-view shapes of the holes **52***a* formed in a substrate of single-crystal silicon with a (110) plane orientation using anisotropic etching. Note that the plan-view shapes of the hole **52***a* are not limited to the example shown in FIG. **4** and may be any shape.

[0038] The pressure chambers C are formed after the formation of the piezoelectric elements **56**.

The pressure chambers C are formed by, for example, performing anisotropic etching on one of the surfaces of the single-crystal silicon substrate having the piezoelectric element **56** already formed therein, the one surface being opposite from the surface where the piezoelectric elements **56** are formed. In this anisotropic etching, for example, a potassium hydroxide (KOH) solution or the like is used as an etching liquid. Also, in a case where an elastic layer **55***a* is formed of silicon oxide, the elastic layer **55***a* functions as a stop layer for stopping the anisotropic etching. After the formation of the pressure chambers C, the flow channel substrate **51** and the like are joined to the pressure chamber substrate **52** with, for example, an adhesive. Note that after the formation of the piezoelectric element **56**, the surface of the single-crystal silicon substrate opposite from the surface where the piezoelectric elements **56** are formed is polished by chemical mechanical polishing (CMP) or the like as needed to flatten the surface and adjust the thickness of the substrate.

[0039] As shown in FIG. **4**, the piezoelectric elements **56** overlap with the pressure chambers C in a plan view. As shown in FIG. **5**, the piezoelectric element **56** has a first electrode **56***a*, a piezoelectric layer **56***b*, and a second electrode **56***c*, and they are laminated in this order in the Z1-direction. Specifically, the first electrode **56***a*, the piezoelectric layer **56***b*, and the second electrode **56***c* are laminated in this order in a lamination direction DL, which is a direction directed from the vibration plate **55** to the piezoelectric element **56**. Note that the piezoelectric element **56** has, in addition to the first electrode **56***a*, the piezoelectric layer **56***b*, and the second electrode **56***c*, a first lead diffusion inhibition layer **56***d* and a second lead diffusion inhibition layer **56***e*, as will be described later based on FIG. **6**.

[0040] The first electrodes **56***a* are individual electrodes disposed for the respective piezoelectric elements **56** and spaced away from one another. Specifically, a plurality of first electrodes **56***a* extending in a direction along the X-axis are arrayed in a direction along the Y-axis with spaces in between. The first electrode **56***a* of each piezoelectric element **56** is supplied with a drive signal including a predetermined voltage pulse from the control unit **20**. Note that details of the first electrode **56***a* will be described later based on FIG. **6**.

[0041] In the example shown in FIGS. **4** and **5**, the piezoelectric layer **56***b* has a belt shape extending in a direction along the Y-axis continuously over the plurality of piezoelectric elements **56**. In the example shown in FIG. **4**, in regions corresponding to the gaps between the adjacent pressure chambers C in a plan view, the piezoelectric layer **56***b* is provided with through-holes **56***b***1** penetrating through the piezoelectric layer **56***b* and extending in a direction along the X-axis. Thus, in a sectional view shown in FIG. **5**, the piezoelectric layer **56***b* is provided individually for each piezoelectric element **56**. Note that the piezoelectric layer **56***b* may be provided individually for the plurality of piezoelectric elements **56**.

[0042] The piezoelectric layer **56***b* is formed of a piezoelectric material containing lead as a constituent element. The piezoelectric material has a perovskite crystal structure expressed by a general composition formula ABO.sub.3 and is, for example, lead zirconate titanate (Pb (Zr, Ti)O.sub.3). Also, in addition to lead (Pb), the piezoelectric layer **56***b* may contain at least one element selected from vanadium (V), niobium (Nb), tantalum (Ta), nickel (Ni), phosphorus (P), arsenic (As), antimony (Sb), and bismuth (Bi). Note that lead contained in the piezoelectric material constituting the piezoelectric layer **56***b* may be an element that does not constitute part of the perovskite crystal structure.

[0043] The second electrode **56***c* is a belt-shaped common electrode extending in a direction along the Y-axis continuously over the plurality of piezoelectric elements **56**. The second electrode **56***c* is supplied with a predetermined constant potential.

[0044] The second electrode **56***c* is formed of, for example, iridium (Ir). Note that a material forming the second electrode **56***c* is not limited to iridium, and may be, for example, a metal material such as titanium (Ti), platinum (Pt), aluminum (Al), nickel (Ni), gold (Au), or copper (Cu) or a conductive oxide such as lanthanum nickel oxide (LaNiO.sub.3: LNO) or strontium ruthenium oxide ($SrRuO_3$: SRO). Also, the second electrode **56***c* may use a single one of these metal materials or may combine two or more of them using a mode such as lamination.

[0045] In the piezoelectric element **56** thus described, upon application of a voltage between the first electrode **56***a* and the second electrode **56***c*, the piezoelectric layer **56***b* deforms due to the inverse piezoelectric effect. The vibration plate **55** is connected to the piezoelectric element **56** and vibrates as the piezoelectric layer **56***b* deforms.

[0046] As shown in FIG. **5**, the vibration plate **55** has the elastic layer **55***a* and an insulating layer **55***b*, which are laminated in this order in the Z1-direction. The insulating layer **55***b* is disposed offset to a position closer to the first electrode **56***a* in the direction of the thickness of the vibration plate **55**.

[0047] The elastic layer **55***a* is a film formed of, for example, silicon oxide (SiO_2). Note that a material forming the elastic layer **55***a* is not limited to SiO_2, and may be, for example, a material containing one element or two or more elements selected from titanium (Ti), silicon (Si), aluminum (Al), tantalum (Ta), chromium (Cr), iridium (Ir), hafnium (Hf), zirconium (Zr), and carbon (C), in an elemental form, an oxide form, or a nitride form.

[0048] The thickness of the elastic layer **55***a* is determined based on, e.g., the thickness and width of the vibration plate **55** and not limited to any particular thickness, but is, for example, in the range of no less than 100 nm and no more than 3000 nm.

[0049] The insulating layer **55***b* is a film formed of, for example, zirconium oxide (ZrO_2) and contains zirconium (Zr). The insulating layer **55***b* thus containing zirconium is disposed offset to a position closer to the first electrode **56***a* in the thickness direction of the vibration plate **55**, which can inhibit bonding of the lead contained in the first electrode **56***a* with the material forming the elastic layer **55***a*. Note that the material forming the insulating layer **55***b* is not limited to ZrO_2 and may be, for example, a material containing one element or two or more elements selected from titanium (Ti), aluminum (Al), tantalum (Ta), chromium (Cr), hafnium (Hf), and zirconium (Zr), in an oxide or nitride form.

[0050] The thickness of the insulating layer **55***b* is determined based on, e.g., the thickness and width of the vibration plate **55** and not limited to any particular thickness, but is, for example, in the range of no less than 100 nm and no more than 2000 nm.

[0051] In the example shown in FIG. **5**, the elastic layer **55***a* and the insulating layer **55***b* are in contact with each other. Note that another layer may be interposed between the elastic layer **55***a* and the insulating layer **55***b*, such as an adhesion layer for enhancing the adhesion between the elastic layer **55***a* and the insulating layer **55***b*. A material forming the adhesion layer is, for example, TiO_X, AlO_X, CrO_X, or TiN. The thickness of the adhesion layer is determined based on the thickness and width of the vibration plate **55** and not limited to any particular thickness, but is, for example, in the range of no less than 20 nm and no more than 2000 nm.

[0052] The elastic layer **55***a* and the insulating layer **55***b* thus described are formed as films in this order on the single-crystal silicon substrate for forming the pressure chamber substrate **52**. For example, in a case where the elastic layer **55***a* is formed of silicon oxide, the elastic layer **55***a* is formed by thermal oxidization of one of the surfaces of the single-crystal silicon substrate. For example, in a case where the insulating layer **55***b* is formed of zirconium oxide, the insulating layer **55***b* is formed by thermal oxidization of a zirconium layer formed by sputtering on the elastic layer **55***a*.

[0053] Note that methods for forming the plurality of films forming the vibration plate **55** are not limited to the examples described above, and any methods may be used. For example, chemical vapor deposition (CVD) or the like may be used to form at least part of the elastic layer **55***a*. Also, in a case where an adhesion layer is provided between the elastic layer **55***a* and the insulating layer **55***b*, the adhesion layer is formed by thermal oxidation of a layer of chromium, titanium, aluminum, or the like formed by sputtering on the elastic layer **55***a*. The thermal oxidation for forming the adhesive layer may be performed concurrently with the thermal oxidation for forming the insulating layer **55***b*. Also, the formation of the adhesive layer is not limited to the method using thermal oxidation and may be performed using, for example, CVD, atomic layer deposition (ALD), or the like.

[0054] The actuator **1** formed by the vibration plate **55** and the piezoelectric element **56** thus described has a vibration region PV that vibrates when the piezoelectric element **56** is driven. The vibration region PV is part of the actuator **1** and is a portion overlapping with the pressure chamber C in a plan view.

[0055] The vibration region PV is divided into an active portion RA and a non-active portion RN. The active portion RA is part of the actuator **1** and is a region overlapping with the pressure chamber C, the first electrode **56***a*, the piezoelectric layer **56***b*, and the second electrode **56***c* when seen in a direction along the Z-axis. The non-active portion RN is part of the actuator **1** and is a region overlapping with the pressure chamber C when seen in a direction along the Z-axis and being different from the active portion RA. The non-active portion RN, when seen in a direction along the Z-axis, is a region where the first electrode **56***a* is not provided and where at least one of the piezoelectric layer **56***b* and the second electrode **56***c* overlaps with the pressure chamber C.

[0056] FIG. **6** is an enlarged view of part VI in FIG. **5**. As shown in FIG. **6**, the first electrode **56***a* has an electrode layer **56***a***1**, a lead-containing layer **56***a***2**, a first mixture layer **56***a***3**, and a second mixture layer **56***a***4**.

[0057] The electrode layer **56***a***1** has, for example, a layer formed of platinum (Pt), a layer formed of iridium (Ir), and a layer formed of titanium (Ti). In other words, the electrode layer **56***a***1** contains platinum, iridium, and titanium. Platinum (Pt) is an electrode material having good conductivity. Thus, using platinum (Pt) as a material forming the first electrode **56***a* can lower the resistance of the first electrode **56***a*. Also, in formation of the piezoelectric layer **56***b*, the layer formed of titanium and island-shaped Ti acts as a crystal nucleus and controls the orientation of the piezoelectric layer **56***b*, enhancing the crystallinity and orientation of the piezoelectric layer **56***b*. Note that in place of or in addition of these layers, a layer formed of a different metal material or conductive oxide may be provided. Also, such a layer may have a layer of a mixture of a plurality of metal materials or a layer formed of an alloy.

[0058] The thickness of the electrode layer **56***a***1** may be, for example, in the range of no less than 90 nm and no more than 160 nm.

[0059] The lead-containing layer **56***a***2** is disposed between the electrode layer **56***a***1** and the vibration plate **55** and contains lead. In the example shown in FIG. **6**, the surface of the lead-containing layer **56***a***2** which faces the Z2-direction forms the surface of the first electrode **56***a* which faces the Z2-direction. In other words, the lead-containing layer **56***a***2** forms the lowermost layer of the first electrode **56***a* in the lamination direction DL. The thickness of the lead-containing layer **56***a***2** is, for example, in the range of no less than 5 nm and no more than 25 nm. Also, the lead-containing layer **56***a***2** is a layer with a higher lead content percentage than the other portions of the electrode layer **56***a***1**. When lead content percentage are measured using, for example, secondary ion mass spectrometry, energy dispersive X-ray analysis, or the like, the lead-containing layer **56***a***2** has a higher lead content percentage than the other portions of the electrode layer **56***a***1**.

[0060] The lead-containing layer **56***a***2** contains a metal element other than lead. For example, the lead-containing layer **56***a***2** contains a material contained in the electrode layer **56***a***1**. Note that the lead-containing layer **56***a***2** may contain a different metal material. Also, the lead contained in the lead-containing layer **56***a***2** may be present in an elemental or alloyed form or in an oxide, nitride, or oxynitride form.

[0061] Of two regions arranged in a crossing direction DC, which is a direction crossing the lamination direction DL, when seen in the lamination direction DL, a region farther from the center of the first electrode **56***a* is a first region RE**1** and a region closer to the center of the first electrode **56***a* is a second region RE**2**. Then, the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** is higher than the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**.

[0062] The size of the first region RE**1** in the crossing direction DC is determined based on, e.g., the thickness and width of the first electrode **56***a* and not limited to any particular size, but is, for example, in the range of no less than 180 nm and no more than 320 nm. In other words, the first region RE**1** is a region extending in the crossing direction DC from the end portion of the first electrode **56***a* toward its center for a distance in the range of no less than 180 nm and no more than 320 nm.

[0063] Although not shown, the first region RE**1** is provided on both sides of the active portion RA in the crossing direction DC. Thus, the second region RE**2** is a region sandwiched by the first regions RE**1** in the crossing direction DC.

[0064] In the example shown in FIG. **6**, the first electrode **56***a* measures approximately 40 μm in width and approximately 110 nm in thickness, and the size of the first region RE**1** in the crossing direction DC is approximately 240 nm. The size of the first region RE**1** may be determined based on the thickness of the first electrode **56***a*. For example, the size of the first region RE**1** is preferably no less than two times and less than four times the thickness of the first electrode **56***a*.

[0065] Note that in the example shown in FIG. **6**, the first electrode **56***a* has a slant surface **56***s* at an end portion in the crossing direction DC, such that its surface facing the Z1-direction slants toward the vibration plate **55**. In this example, the slant surface **56***s* is located in the first region RE**1**.

[0066] When lead content percentages are measured using, for example, secondary ion mass spectrometry, energy dispersive X-ray analysis, or the like, the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** is higher than the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**. A lead content percentage is calculated from, for example, the ratio of a lead content relative to the content of a different material contained in the electrode layer **56***a***1**.

[0067] When lead content percentages in the lead-containing layer **56***a***2** are measured using, for example, secondary ion mass spectrometry, contents of two or more elements including lead from the elements contained in the lead-containing layer **56***a***2** are measured over a particular area in a plan view. Then, the content of lead is standardized by the sum of the contents of lead and the other element. The lead content percentage is thus calculated. Note that in a case of measuring the content of oxygen as an element other than lead, the lead content percentage may be calculated by standardization of the lead content by the oxygen content. Also, a lead content percentage may be calculated by standardization of a lead content by the content of a particular material.

[0068] Note that in a case of calculating lead content percentages in the lead-containing layer **56***a***2** in the first region RE**1** and the second region RE**2**, the measurements are preferably conducted at the center position of the lead-containing layer **56***a***2** in the first region RE**1** and the center position of the lead-containing layer **56***a***2** in the second region RE**2** in terms of the lamination direction DL and the crossing direction DC. Also, the measurements are preferably conducted at the same height in the lamination direction DL.

[0069] Note that a lead content percentage may be calculated using the average of a plurality of lead content percentages at a plurality of locations on the lead-containing layer **56***a***2** in the lamination direction DL or the crossing direction DC.

[0070] The first region RE**1** and the second region RE**2** are included in the active portion RA. In the example shown in FIG. **6**, the active portion RA is divided into the first region RE**1** and the second region RE**2**, and the first region RE**1** and the second region RE**2** are adjacent to each other. Also, the first region RE**1** is adjacent to the non-active portion RN. The second region RE**2** is a region located toward the center of the first electrode **56***a* when seen in the lamination direction DL. The first region RE**1** is a region located more outward than the second region RE**2** when seen in the lamination direction DL.

[0071] In this way, the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** is higher than the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**. In other words, the lead content in the center portion of the first electrode **56***a* is lower than the lead content in the end portion of the first electrode **56***a*. This enables electrical resistance in the center portion of the first electrode **56***a* to be smaller than electrical resistance in the end portion of the first electrode **56***a*. Also, the lead in the lead-containing layer **56***a***2** in the first region RE**1** can reduce intrusion of the lead in the piezoelectric layer **56***b* into the lead-containing layer **56***a***2** in the second region RE**2** through the lead-containing layer **56***a***2** in the first region RE**1**. This as a result allows the piezoelectric element **56** to have an adequate region functioning as the active portion RA.
[0072] Due to the aforementioned relation concerning the lead-containing layer **56***a***2** in the first region RE**1** and the second region RE**2**, as the average of the entire first electrode **56***a*, the lead content percentage in the first electrode **56***a* in the first region RE**1** is higher than the lead content percentage in the first electrode **56***a* in the second region RE**2**. A high lead content percentage, i.e., a large lead content, means a high electrical resistance and a large voltage drop.
[0073] Hence, a voltage drop in the first electrode **56***a* in the first region RE**1** is larger than a voltage drop in the first electrode **56***a* in the second region RE**2**. Thus, placing the first region RE**1** at the border between the active portion RA and the non-active portion RN of the piezoelectric element **56** enables the displacement amount of the piezoelectric element **56** to be smaller and smaller from the active portion RA toward the non-active portion RN of the piezoelectric element **56**. This can mitigate the stress generated between the active portion RA and the non-active portion RN of the piezoelectric element **56**.
[0074] Further, the interface between the first electrode **56***a* and the vibration plate **55** in the first region RE**1** or the interface between the first electrode **56***a* and the second lead diffusion inhibition layer **56***e* to be described later in the first region RE**1** can be roughened. This can improve the adhesion between the first electrode **56***a* and the vibration plate **55** or the second lead diffusion inhibition layer **56***e* in the first region RE**1** due to the anchoring effect. This as a result favorably prevents the first electrode **56***a* in the first region RE**1** from being peeled off the vibration plate **55**.
[0075] A lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** preferably decreases toward the second region RE**2**. This makes a voltage drop smaller toward the center portion of the first electrode **56***a* and favorably ensures that the piezoelectric element **56** has an adequate region functioning as the active portion RA. This also advantageously makes it easier to mitigate stress generated between the active portion RA and the non-active portion RN of the piezoelectric element **56**.
[0076] The ratio of the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** to the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2** is preferably 1.01 or greater. Specifically, α/β is preferably 1.01 or greater, where a is a lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** and B is a lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**. This makes it possible to favorably obtain the effect achieved when the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** is higher than the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**.
[0077] In the example shown in FIG. **6**, the thickness of the lead-containing layer **56***a***2** in the second region RE**2** is substantially constant, whereas the thickness of the lead-containing layer **56***a***2** in the first region RE**1** becomes thinner with distance from the second region RE**2**. Note that the thickness of the lead-containing layer **56***a***2** in the first region RE**1** may be substantially constant, similarly to the thickness of the lead-containing layer **56***a***2** in the second region RE**2**.
[0078] The lead-containing layer **56***a***2** may contain at least one element selected from vanadium (V), niobium (Nb), tantalum (Ta), nickel (Ni), phosphorus (P), arsenic (As), antimony (Sb), and bismuth (Bi). For example, in a case where the piezoelectric layer **56***b* contains at least one element

selected from V, Nb, Ta, Ni, P, As, Sb, and Bi, a content percentage of the at least one element in the lead-containing layer **56***a***2** in the first region RE**1** is preferably higher than a content percentage of the at least one element in the lead-containing layer **56***a***2** in the second region RE**2**.

[0079] In a case where the piezoelectric layer **56***b* contains at least one element selected from V, Nb, Ta, Ni, P, As, Sb, and Bi, oxygen deficit in the piezoelectric layer **56***b* can be mitigated. As a result, the drive speed for the piezoelectric element **56** can be increased. Also, setting a higher content percentage of the above element for the lead-containing layer **56***a***2** in the first region RE**1** than for the lead-containing layer **56***a***2** in the second region RE**2** can offer the same effect achieved by setting a higher lead content percentage for the lead-containing layer **56***a***2** in the first region RE**1** than for the lead-containing layer **56***a***2** in the second region RE**2**.

[0080] The thickness of the lead-containing layer **56***a***2** is not limited to any particular thickness, but is, for example, in the range of no less than 5 nm and no more than 25 nm. The lead-containing layer **56***a***2** may be a closely packed layer or may be a sparse layer where lead in an elemental or alloyed form or in an oxide, nitride, or oxynitride form is scattered in the shapes of island.

[0081] The first mixture layer **56***a***3** is disposed between the electrode layer **56***a***1** and the lead-containing layer **56***a***2** in the first region RE**1** and is formed of a mixed material of lead and the material forming the electrode layer **56***a***1**. The provision of the first mixture layer **56***a***3** enables further improvement in the adhesion between the first electrode **56***a* and the vibration plate **55** in the first region RE**1**. This as a result can inhibit peeling of the first electrode **56***a* off the vibration plate **55** or the occurrence of cracking. Note that, for illustrative convenience, FIG. **6** distinctly depicts the interface between the first mixture layer **56***a***3** and the electrode layer **56***a***1** and the interface between the first mixture layer **56***a***3** and the lead-containing layer **56***a***2**, but these interfaces do not have to be distinct.

[0082] A thickness t**1** of the first mixture layer **56***a***3** is not limited to any particular thickness, but is, for example, in the range of no less than 5 nm and no more than 30 nm.

[0083] The second mixture layer **56***a***4** is disposed between the electrode layer **56***a***1** and the lead-containing layer **56***a***2** in the second region RE**2** and is formed of a mixed material of lead and the material forming the electrode layer **56***a***1**. In the example shown in FIG. **6**, the second mixture layer **56***a***4** is integral with the first mixture layer **56***a***3** as the same layer. Note that the second mixture layer **56***a***4** may be separate from the first mixture layer **56***a***3**. Also, for illustrative convenience, FIG. **6** distinctly depicts the interface between the second mixture layer **56***a***4** and the electrode layer **56***a***1** and the interface between the first mixture layer **56***a***3** and the lead-containing layer **56***a***2**, but these interfaces do not have to be distinct.

[0084] A thickness t**2** of the second mixture layer **56***a***4** is preferably smaller than the thickness t**1** of the first mixture layer **56***a***3**. Thus, the thickness t**1** of the first mixture layer **56***a***3** is preferably larger than the thickness t**2** of the second mixture layer **56***a***4**. This can favorably inhibit both of the occurrence of cracking in the first region RE**1** and peeling of the first electrode **56***a* off the vibration plate **55**.

[0085] The thickness t**2** of the second mixture layer **56***a***4** is not limited to any particular thickness, but is, for example, in the range of no less than 5 nm and no more than 30 nm.

[0086] Note that the first mixture layer **56***a***3** and the second mixture layer **56***a***4** are layers with lower lead content percentages than the lead-containing layer **56***a***2**. The lead content percentages in the first mixture layer **56***a***3** and the second mixture layer **56***a***4** are measured similarly to the lead content percentages described above.

[0087] For example, the first electrode **56***a* is formed after the formation of the vibration plate **55**, using a film formation technique such as sputtering and processing techniques such as photolithography and etching. Also, lead may be contained into the lead-containing layer **56***a***2** by, for example, introduction of lead through ion implantation or film formation of a material containing lead. Note that in order for the lead-containing layer **56***a***2** to have different lead content percentages in the first region RE**1** and the second region RE**2**, the amount of lead introduced for ion implantation may be changed, or the film formation conditions and target may be changed for each region.

[0088] The piezoelectric element **56** has the first lead diffusion inhibition layer **56***d* and the second lead diffusion inhibition layer **56***e* in addition to the first electrode **56***a*, the piezoelectric layer **56***b*, and the second electrode **56***c* described above.

[0089] The first lead diffusion inhibition layer **56***d* is disposed between the first electrode **56***a* and the piezoelectric layer **56***b* at least in the second region RE**2** and inhibits diffusion of lead from the piezoelectric layer **56***b* to the first electrode **56***a*. This can inhibit entry of the lead from the piezoelectric layer **56***b* to the first electrode **56***a* in the second region RE**2**. The first lead diffusion inhibition layer **56***d* contains titanium. Note that the first lead diffusion inhibition layer **56***d* may contain a metal element other than titanium. Also, titanium contained in the first lead diffusion inhibition layer **56***d* may be present in an elemental or alloyed form or may be present in an oxide, nitride, or oxynitride form.

[0090] In the example shown in FIG. **6**, the first lead diffusion inhibition layer **56***d* is provided over an area including the first region RE**1**, the second region RE**2**, and a third region RE**3**, the third region RE**3** being a region connected to the first region RE**1** and opposite from the second region RE**2** in the crossing direction DC when seen in the lamination direction DL. Note that the third region RE**3** is included in the non-active portion RN.

[0091] The size of the third region RE**3** in the crossing direction DC is determined based on, e.g., the thickness and width of the first electrode **56***a* and not limited to any particular size, but is, for example, in the range of no less than 90 nm and no more than 320 nm. Thus, the third region RE**3** is a region extending from the end portion of the first electrode **56***a* outward in the crossing direction DC for a distance in the range of no less than 90 nm and no more than 320 nm. In the example shown in FIG. **6**, the size of the third region RE**3** in the crossing direction DC is approximately 240 nm.

[0092] The titanium content percentage in the first lead diffusion inhibition layer **56***d* in the third region RE**3** is preferably higher than the titanium content percentage in the first lead diffusion inhibition layer **56***d* in the first region RE**1**. When titanium content percentages are measured using, for example, secondary ion mass spectrometry, energy dispersive X-ray analysis, or the like, the titanium content percentage in the first lead diffusion inhibition layer **56***d* in the third region RE**3** is higher than the titanium content percentage in the first lead diffusion inhibition layer **56***d* in the second region RE**2**. A titanium content percentage is calculated from, for example, the ratio of a titanium content relative to the content of another material contained in the first lead diffusion inhibition layer **56***d*. Thus, lead escaping from the piezoelectric layer **56***b* in third region RE**3** bonds preferentially with the titanium in the first lead diffusion inhibition layer **56***d* in the third region RE**3**, which can inhibit entry of the lead to the first region RE**1** and the second region RE**2**.

[0093] Note that the measurement of the titanium content percentages is preferably conducted at the center position of the first lead diffusion inhibition layer **56***d* in the first region RE**1** and the center position of the first lead diffusion inhibition layer **56***d* in the third region RE**3** in terms of the lamination direction DL and the crossing direction DC.

[0094] Also, the first lead diffusion inhibition layer **56***d* has the function of controlling the orientation of the piezoelectric layer **56***b*. The piezoelectric layer **56***b* is in contact with the first lead diffusion inhibition layer **56***d* in the first region RE**1** and the second region RE**2**. This enables the first lead diffusion inhibition layer **56***d* to control the orientation of the piezoelectric layer **56***b* in the first region RE**1** and the second region RE**2**.

[0095] As described earlier, the thickness of the lead-containing layer **56***a***2** in the first region RE**1** becomes thinner with distance from the second region RE**2**. Thus, the surface of the first lead diffusion inhibition layer **56***d* which is opposite from the vibration plate **55** slants relative to the plate surface of the vibration plate **55**, and the surface of the first electrode **56***a* which is opposite from the vibration plate **55** slants relative to the plate surface of the vibration plate **55** as well.

[0096] The first lead diffusion inhibition layer **56***d* covers the first electrode **56***a* in the first region RE**1**. In the first region RE**1**, an angle θ**1** formed between the surface of the first lead diffusion inhibition layer **56***d* which is opposite from the vibration plate **55** and the plate surface of the vibration plate **55** is smaller than an angle θ**2** formed between the surface of the first electrode **56***a* which is opposite from the vibration plate **55** and the plate surface of the vibration plate **55**. The first lead diffusion inhibition layer **56***d* becomes thicker and thicker from the first region RE**1** to the non-active portion RN. This makes it possible to reduce the difference between the ability of the first lead diffusion inhibition layer **56***d* to control the orientation of the piezoelectric material in the first region RE**1** and the ability of the first lead diffusion inhibition layer **56***d* to control the orientation of the piezoelectric material in the second region RE**2**. This enables enhancement of the orientation of the piezoelectric layer **56***b* in the active portion RA.
[0097] In the first region RE**1**, the angle θ**1** formed between the surface of the first electrode **56***a* which is opposite from the vibration plate **55** and the plate surface of the vibration plate **55** is preferably no less than 10 degrees and less than 60 degrees. This can not only make the liquid ejection head **50** compact in size, but also inhibit entry of lead into the first electrode **56***a* in the first region RE**1**.
[0098] The thickness of the first lead diffusion inhibition layer **56***d* is not limited to any particular thickness, but is, for example, in the range of no less than 20 nm and no more than 80 nm. Note that the first lead diffusion inhibition layer **56***d* is provided as needed and may be omitted.
[0099] The first lead diffusion inhibition layer **56***d* is formed using, for example, a film formation technique such as sputtering. In order for the first lead diffusion inhibition layer **56***d* to have different titanium content percentages in the first region RE**1** and the third region RE**3**, the film formation conditions and target may be changed for each region.
[0100] The second lead diffusion inhibition layer **56***e* is disposed between the first electrode **56***a* and the vibration plate **55** in each of the first region RE**1** and the second region RE**2** and inhibits diffusion of lead from the first electrode **56***a* to the vibration plate **55**. The second lead diffusion inhibition layer **56***e* contains titanium. The second lead diffusion inhibition layer **56***e* may contain a metal element other than titanium. Also, the titanium contained in the second lead diffusion inhibition layer **56***e* may be present in an elemental or alloyed form or in an oxide, nitride, or oxynitride form.
[0101] The titanium content percentage in the second lead diffusion inhibition layer **56***e* in the first region RE**1** is preferably higher than the titanium content percentage in the second lead diffusion inhibition layer **56***e* in the second region RE**2**. When lead contents are measured using, for example, secondary ion mass spectrometry, energy dispersive X-ray analysis, or the like, the titanium content percentage in the second lead diffusion inhibition layer **56***e* in the first region RE**1** is higher than the titanium content percentage in the second lead diffusion inhibition layer **56***e* in the second region RE**2**. For example, a titanium content percentage is calculated from the ratio of a titanium content relative to the content of another material contained in the second lead diffusion inhibition layer **56***e*.
[0102] The titanium in the second lead diffusion inhibition layer **56***e* forms a compound such as TiPbO.sub.x by bonding with the lead in the lead-containing layer **56***a***2**. Thus, setting a higher titanium content percentage for the lead-containing layer **56***a***2** in the first region RE**1** than for the lead-containing layer **56***a***2** in the second region RE**2** can inhibit entry of lead from the first region RE**1** to the second region RE**2**.
[0103] A titanium oxide content percentage in the second lead diffusion inhibition layer **56***e* in the first region RE**1** is preferably higher than a titanium oxide content percentage in the second lead diffusion inhibition layer **56***e* in the second region RE**2**. This makes the degree of oxidation of the lead-containing layer **56***a***2** in the first region RE**1** higher than the degree of oxidation in the lead-containing layer **56***a***2** in the second region RE**2** and can inhibit oxygen loss from the piezoelectric layer **56***b* through the first region RE**1**.

[0104] The content percentage of TiO_2 in the second lead diffusion inhibition layer **56***e* in the first region RE**1** is preferably higher than the content percentage of TiO_2 in the second lead diffusion inhibition layer **56***e* in the second region RE**2**. TiO_2 is chemically stable compared to titanium oxides of other oxidation numbers. Hence, setting a higher content percentage of TiO_2 for the lead-containing layer **56***a***2** in the first region RE**1** than for the lead-containing layer **56***a***2** in the second region RE**2** can favorably inhibit oxygen loss from the piezoelectric layer **56***b* through the first region RE**1**.
[0105] The thickness of the second lead diffusion inhibition layer **56***e* is not limited to any particular thickness, but is, for example, in the range of no less than 1 nm and no more than 10 nm. Note that the second lead diffusion inhibition layer **56***e* is provided as needed and may be omitted.
[0106] The second lead diffusion inhibition layer **56***e* is formed using, for example, a film formation technique such as sputtering. Also, titanium or titanium oxide may be contained into the second lead diffusion inhibition layer **56***e* by, for example, introduction of titanium through ion implantation or film formation of a material containing titanium or titanium oxide. In order for the second lead diffusion inhibition layer **56***e* to have different content percentages of titanium or titanium oxide in the first region RE**1** and the second region RE**2**, the film formation conditions and target may be changed for each region.
[0107] FIG. **7** is a diagram showing the relations between lead content information on the lead-containing layer **56***a***2** and evaluations of the liquid ejection head **50**. FIG. **7** shows evaluation results on the lifetime and electrical resistance of the liquid ejection head **50** regarding samples No. 1 to No. 4 whose lead content percentages are different from one another in one or both of the first region RE**1** and the second region RE**2**. Note that the lead content percentages in the lead-containing layer **56***a***2** are measured by, for example, secondary ion mass spectrometry. A lead content percentage in each of the first region RE**1** and the second region RE**2** is calculated by measuring an oxygen content and a lead content over a region of an area of 1 mm^2 in a plan view at measurement time intervals and standardizing the lead content by the oxygen content.
[0108] In sample No. 1, a lead content percentage a in the lead-containing layer **56***a***2** in the first region RE**1** is 20 atm % or higher, and a lead content percentage β in the lead-containing layer **56***a***2** in the second region RE**2** is 1.7 atm %. In this sample, α/β is greater than 11.
[0109] In sample No. 2, the lead content percentage a in the lead-containing layer **56***a***2** in the first region RE**1** is 20 atm % or higher, and the lead content percentage β in the lead-containing layer **56***a***2** in the second region RE**2** is 18.7 atm %. In this sample, α/β is greater than 1.07.
[0110] In sample No. 3, the lead content percentage a in the lead-containing layer **56***a***2** in the first region RE**1** is 1.5 atm % or lower, and the lead content percentage β in the lead-containing layer **56***a***2** in the second region RE**2** is 1.7 atm %. In this sample, α/β is smaller than 0.88.
[0111] In sample No. 4, the lead content percentage a in the lead-containing layer **56***a***2** in the first region RE**1** is 1.5 atm % or lower, and the lead content percentage β in the lead-containing layer **56***a***2** in the second region RE**2** is 18.7 atm %. In this sample, α/β is smaller than 0.08.
[0112] Regarding the lifetime evaluation in FIG. **7**, “P” indicates that the active portion RA is unlikely to break, and “F” indicates that the active portion RA is likely to break. Also, regarding the electrical resistance evaluation in FIG. **7**, “A” indicates that the effect of lowering the electrical resistance in the piezoelectric element **56** is notable, and “B” indicates that the effect of lowering the electrical resistance in the piezoelectric element **56** is observed, but only slightly. As to the overall evaluation in FIG. **7**, “A,” “B,” “C,” and “D” indicate ratings of the overall evaluation of lifetime and electrical resistance, “A” being the highest rating and “D” being the lowest rating.
[0113] As shown in FIG. **7**, the active portion RA is unlikely to break in Samples No. 1 and No. 2, whereas the active portion RA is likely to break in Samples No. 3 and No. 4.
[0114] Also, the effect of lowering the electrical resistance in the piezoelectric element **56** is notable in samples No. 1 and No. 3 and is only slightly observed in samples No. 2 and No. 4.
[0115] The order of ratings of the overall evaluation are as follows: Sample No. 1, Sample No. 2,

Sample No. 3, and Sample No. 4, with Sample No. 1 being the highest rating.

[0116] As is understood from the above, the occurrence of cracking can be favorably inhibited when the lead content percentage in the lead-containing layer **56***a***2** in the first region RE**1** is higher than the lead content percentage in the lead-containing layer **56***a***2** in the second region RE**2**.

2. Modifications

[0117] The modes exemplified above can be variously modified. Specific modifications that may be applied to the above-described modes are exemplified below. Note that any of two or more modes selected from the following examples may be combined as needed without conflicting with each other.

2-1. Modification 1

[0118] In the modes described above, the piezoelectric layer **56***b* is provided commonly for the plurality of pressure chambers C; however, the present disclosure is not limited to this, and the piezoelectric layer **56***b* may be divided for the pressure chambers C. Also, the second electrode **56***c* may be individual electrodes, in which case, the first electrode **56***a* may be a common electrode, or the first electrode **56***a* and the second electrode **56***c* may both be individual electrodes.

2-2. Modification 2

[0119] In the modes described above, the serial-type liquid ejection apparatus **100** in which the carriage **41** having the liquid ejection head **50** mounted thereto reciprocates is used as an example; however, the present disclosure can also be applied to a line-type liquid ejection apparatus in which the plurality of nozzles N are distributed over the entire width of the medium M.

2-3. Modification 3

[0120] The liquid ejection apparatus **100** exemplified in the above modes may be employed not only for a device dedicated for printing, but also various other devices such as a facsimile device or a copier device. However, the purpose of the liquid ejection apparatus of the present disclosure is not limited to printing. For example, a liquid ejection apparatus that ejects a solution of a color material may be used as a manufacturing apparatus that produces color filters for liquid crystal display devices. Also, a liquid ejection apparatus that ejects a solution of a conductive material may be used as a manufacturing apparatus that produces wiring and electrodes on wiring boards.

3. The Overview of the Present Disclosure

[0121] The following supplementarily provides the overview of the present disclosure.

[0122] (Supplemental Note 1) A first mode as a preferred example of a liquid ejection head of the present disclosure includes: a piezoelectric element having a first electrode, a piezoelectric layer, and a second electrode and a vibration plate connected to the piezoelectric element, the first electrode, the piezoelectric layer, and the second electrode are laminated in this order in a lamination direction which is a direction directed from the vibration plate to the piezoelectric element, the piezoelectric layer is formed of a piezoelectric material containing lead as a constituent element, the first electrode has an electrode layer and a lead-containing layer which is disposed between the electrode layer and the vibration plate and contains lead, and when, of two regions arranged in a crossing direction, which is a direction crossing the lamination direction, when seen in the lamination direction, a region farther away from a center of the first electrode is a first region and a region closer to the center of the first electrode is a second region, a first lead content percentage in the lead-containing layer in the first region is higher than a second lead content percentage in the lead-containing layer in the second region.

[0123] In the above mode, the first lead content percentage in the lead-containing layer in the first region is higher than the second lead content percentage in the lead-containing layer in the second region. In other words, the lead content in the center portion of the first electrode is smaller than the lead content in the end portion of the first electrode. This enables the electrical resistance in the center portion of the first electrode to be smaller than the electrical resistance in the end portion of the first electrode. Moreover, the lead in the lead-containing layer in the first region can reduce intrusion of lead from the piezoelectric layer to the second region. This as a result ensures that the

piezoelectric element has an adequate region functioning as an active portion.
[0124] Also, a voltage drop in the first electrode in the first region can be made larger than a voltage drop in the first electrode in the second region. Thus, placing the first region at the border between the active portion and the non-active region of the piezoelectric element enables the displacement amount of the piezoelectric element to be smaller and smaller from the active portion to the non-active portion of the piezoelectric element. This enables mitigation of stress generated between the active and non-active portions of the piezoelectric element.
[0125] Further, the interface between the first electrode and the vibration plate in the first region can be roughened. This enables improvement of the adhesion between the first electrode and the vibration plate in the first region due to the anchoring effect. This as a result can favorably prevent peeling of the first electrode off the vibration plate.
[0126] (Supplementary Note 2) In a second mode as a preferred example of the first mode, the first lead content percentage in the lead-containing layer in the first region decreases toward the second region. The above mode can favorably ensure that the piezoelectric element has an adequate region functioning as an active portion. The above mode also advantageously mitigate stress generated between the active and non-active portions of the piezoelectric element.
[0127] (Supplementary Note 3) In a third mode as a preferred example of the first or second mode, a ratio of the first lead content percentage in the lead-containing layer in the first region to the second lead content percentage in the lead-containing layer in the second region is 1.01 or greater. The above mode favorably offers the effect achieved by setting a higher lead content percentage for the lead-containing layer in the first region than for the lead-containing layer in the second region.
[0128] (Supplementary Note 4) In a fourth mode as a preferred mode of any one of the first to third modes, the first electrode has a first mixture layer which is disposed between the electrode layer and the lead-containing layer in the first region and is formed of a mixed material of lead and a material forming the electrode layer. The above mode can further improve the adhesion between the first electrode and the vibration plate in the first region. This as a result can inhibit the occurrence of cracking in the first region.
[0129] (Supplementary Note 5) In a fifth mode as a preferred example of any one of the first to fourth modes, the first electrode has a second mixture layer which is disposed between the electrode layer and the lead-containing layer in the second region and is formed of a mixed material of lead and a material forming the electrode layer, and thickness of the first mixture layer is thicker than thickness of the second mixture layer. The above mode can favorably inhibit both the occurrence of cracking in the first region and peeling of the first electrode off the vibration plate.
[0130] (Supplementary Note 6) In a sixth mode as a preferred example of any one of the first to fifth modes, the piezoelectric element further has a first lead diffusion inhibition layer which is disposed between the first electrode and the piezoelectric layer at least in the second region and which inhibits diffusion of lead from the piezoelectric layer to the first electrode. The above mode can inhibit entry of lead from the piezoelectric layer to the first electrode in the second region.
[0131] (Supplementary Note 7) In a seventh mode as a preferred example of any one of the first to sixth modes, the piezoelectric element further has a second lead diffusion inhibition layer which is disposed between the first electrode and the vibration plate in each of the first region and the second region and which inhibits diffusion of lead from the first electrode to the vibration plate, the second lead diffusion inhibition layer contains titanium, and a first titanium content percentage in the second lead diffusion inhibition layer in the first region is higher than a second titanium content percentage in the second lead diffusion inhibition layer in the second region. In the above mode, titanium in the lead-containing layer bonds with lead, forming a compound such as TiPbO.sub.x. Thus, setting a higher titanium content percentage for the lead-containing layer in the first region than for the lead-containing layer in the second region allows the lead in the piezoelectric layer to enter the second region through the first region.
[0132] (Supplementary Note 8) In an eighth mode as a preferred example of the seventh mode, a

first titanium oxide content percentage in the second lead diffusion inhibition layer in the first region is higher than a second titanium oxide content percentage in the second lead diffusion inhibition layer in the second region. In the above mode, because the degree of oxidation of the lead-containing layer in the first region is higher than the degree of oxidation of the lead-containing layer in the second region, oxygen loss from the piezoelectric layer through the first region can be inhibited.

[0133] (Supplementary Note 9) In a ninth mode as a preferred example of the eighth mode, the second lead diffusion inhibition layer contains TiO.sub.2, and a first TiO.sub.2 content percentage in the second lead diffusion inhibition layer in the first region is higher than a second TiO.sub.2 content percentage in the second lead diffusion inhibition layer in the second region. In the above mode, because TiO.sub.2 is chemically stable compared to titanium oxides of other oxidation numbers, setting a higher TiO.sub.2 content percentage for the lead-containing layer in the first region than for the lead-containing layer in the second region can favorably inhibit oxygen loss from the piezoelectric layer through the first region.

[0134] (Supplementary Note 10) In a tenth mode as a preferred example of any one of the sixth to ninth modes, the first lead diffusion inhibition layer is provided over a range including the first region, the second region, and a third region which is, when seen in the lamination direction, a region opposite from the second region in the crossing direction and connected to the first region, and a third titanium content percentage in the first lead diffusion inhibition layer in the third region is higher than a first titanium content percentage in the first lead diffusion inhibition layer in the first region. In the above mode, because lead escaping from the piezoelectric layer in the third region bonds preferentially with titanium, entry of lead into the first and second regions can be inhibited.

[0135] (Supplementary Note 11) In an 11th mode as a preferred example of any one of the sixth to tenth modes, the first lead diffusion inhibition layer covers the first electrode in the first region, and in the first region, an angle formed between a surface of the first lead diffusion inhibition layer which is opposite from the vibration plate and a plate surface of the vibration plate is smaller than an angle formed between a surface of the first electrode which is opposite from the vibration plate and the plate surface of the vibration plate. The above mode can reduce the difference between the ability of the first lead diffusion inhibition layer to control the orientation of the piezoelectric material in the first region and the ability of the first lead diffusion inhibition layer to control the orientation of the piezoelectric material in the second region. This can enhance the orientation of the piezoelectric layer.

[0136] (Supplementary Note 12) In a 12th mode as a preferred example of the 11th mode, the first lead diffusion inhibition layer has a function of controlling orientation of the piezoelectric layer, and the piezoelectric layer is in contact with the first lead diffusion inhibition layer over the first region and the second region. In the above mode, the first lead diffusion inhibition layer can control the orientation of the piezoelectric layer in the first region and the second region.

[0137] (Supplementary Note 13) In a 13th mode as a preferred example of any one of the first to 12th modes, the vibration plate has an insulating layer containing zirconium, and the insulating layer is disposed offset to a location closer to the first electrode in a direction of thickness of the vibration plate. The above mode can inhibit bonding of the lead contained in the first electrode with the material forming the vibration plate.

[0138] (Supplementary Note 14) In a 14th mode as a preferred example of any one of the first to 13th modes, the piezoelectric layer contains at least one element selected from V, Nb, Ta, Ni, P, As, Sb, and Bi, and a first content percentage of the at least one element in the lead-containing layer in the first region is higher than a second content percentage of the at least one element in the lead-containing layer in the second region. In the above mode, when the piezoelectric layer contains at least one element selected from V, Nb, Ta, Ni, P, As, Sb, and Bi, oxygen deficit in the piezoelectric layer can be mitigated. As a result, drive speed for the piezoelectric element can be increased. Also, setting a higher content percentage of the above element for the lead-containing layer in the first region than for the lead-containing layer in the second region can offer an effect similar to the effect achieved by setting a higher lead content percentage for the lead-containing layer in the first region than for the lead-containing layer in the second region.

[0139] (Supplementary Note 15) In a 15th mode as a preferred example of any one of the first to 14th modes, in the first region, an angle formed between a surface of the first electrode which is opposite from the vibration plate and a plate surface of the vibration plate is no less than 10 degrees and less than 60 degrees. The above mode can not only make the liquid ejection head compact in size, but also inhibit entry of lead into the second region.

Claims

1. A liquid ejection head comprising: a piezoelectric element having a first electrode, a piezoelectric layer, and a second electrode and a vibration plate, wherein the first electrode, the piezoelectric layer, and the second electrode are laminated in this order in a lamination direction which is a direction directed from the vibration plate to the piezoelectric element, the piezoelectric layer is formed of a piezoelectric material containing lead as a constituent element, the first electrode has an electrode layer and a lead-containing layer which is disposed between the electrode layer and the vibration plate and contains lead, and when, of two regions arranged in a crossing direction, which is a direction crossing the lamination direction, when seen in the lamination direction, a region farther away from a center of the first electrode is a first region and a region closer to the center of the first electrode is a second region, a first lead content percentage in the lead-containing layer in the first region is higher than a second lead content percentage in the lead-containing layer in the second region.

2. The liquid ejection head according to claim 1, wherein the first lead content percentage in the lead-containing layer in the first region decreases toward the second region.

3. The liquid ejection head according to claim 1, wherein a ratio of the first lead content percentage in the lead-containing layer in the first region to the second lead content percentage in the lead-containing layer in the second region is 1.01 or greater.

4. The liquid ejection head according to claim 1, wherein the first electrode has a first mixture layer which is disposed between the electrode layer and the lead-containing layer in the first region and is formed of a mixed material of lead and a material forming the electrode layer.

5. The liquid ejection head according to claim 4, wherein the first electrode has a second mixture layer which is disposed between the electrode layer and the lead-containing layer in the second region and is formed of a mixed material of lead and a material forming the electrode layer, and thickness of the first mixture layer is thicker than thickness of the second mixture layer.

6. The liquid ejection head according to claim 1, wherein the piezoelectric element further has a first lead diffusion inhibition layer which is disposed between the first electrode and the piezoelectric layer at least in the second region and which inhibits diffusion of lead from the piezoelectric layer to the first electrode.

7. The liquid ejection head according to claim 1, wherein the piezoelectric element further has a second lead diffusion inhibition layer which is disposed between the first electrode and the vibration plate in each of the first region and the second region and which inhibits diffusion of lead from the first electrode to the vibration plate, the second lead diffusion inhibition layer contains titanium, and a first titanium content percentage in the second lead diffusion inhibition layer in the first region is higher than a second titanium content percentage in the second lead diffusion inhibition layer in the second region.

8. The liquid ejection head according to claim 7, wherein a first titanium oxide content percentage in the second lead diffusion inhibition layer in the first region is higher than a second titanium oxide content percentage in the second lead diffusion inhibition layer in the second region.

9. The liquid ejection head according to claim 8, wherein the second lead diffusion inhibition layer contains TiO_2, and a first TiO_2 content percentage in the second lead diffusion inhibition layer in the first region is higher than a second TiO_2 content percentage in the second lead diffusion inhibition layer in the second region.
10. The liquid ejection head according to claim 6, wherein the first lead diffusion inhibition layer is provided over a range including the first region, the second region, and a third region which is, when seen in the lamination direction, a region opposite from the second region in the crossing direction and connected to the first region, and a third titanium content percentage in the first lead diffusion inhibition layer in the third region is higher than a first titanium content percentage in the first lead diffusion inhibition layer in the first region.
11. The liquid ejection head according to claim 6, wherein the first lead diffusion inhibition layer covers the first electrode in the first region, and in the first region, an angle formed between a surface of the first lead diffusion inhibition layer which is opposite from the vibration plate and a plate surface of the vibration plate is smaller than an angle formed between a surface of the first electrode which is opposite from the vibration plate and the plate surface of the vibration plate.
12. The liquid ejection head according to claim 11, wherein the first lead diffusion inhibition layer has a function of controlling orientation of the piezoelectric layer, and the piezoelectric layer is in contact with the first lead diffusion inhibition layer over the first region and the second region.
13. The liquid ejection head according to claim 1, wherein the vibration plate has an insulating layer containing zirconium, and the insulating layer is disposed offset to a location closer to the first electrode in a direction of thickness of the vibration plate.
14. The liquid ejection head according to claim 1, wherein the piezoelectric layer contains at least one element selected from V, Nb, Ta, Ni, P, As, Sb, and Bi, and a first content percentage of the at least one element in the lead-containing layer in the first region is higher than a second content percentage of the at least one element in the lead-containing layer in the second region.
15. The liquid ejection head according to claim 1, wherein in the first region, an angle formed between a surface of the first electrode which is opposite from the vibration plate and a plate surface of the vibration plate is no less than 10 degrees and less than 60 degrees.